Volume 26

Sentinel Endpoint Intelligence Platform (SEIP)

Version 1.0

Vision

Project Sentinel shall continuously understand, verify and improve the security posture of every authorized endpoint.

An endpoint includes:

- Windows
- Linux
- macOS
- Servers
- Workstations
- Virtual Machines
- Containers
- Edge Devices

Future:

- IoT
- Industrial Devices
- Medical Devices
- OT Networks

Philosophy

Traditional Endpoint Security asks

> Is malware running?

Project Sentinel asks

> Do I completely understand this endpoint?

That changes everything.

Endpoint Digital Twin

Every endpoint receives its own Digital Twin.

Example

```
Endpoint
```

The endpoint becomes a living object.

Chapter 1

Endpoint Identity Engine

Every endpoint receives

Unique Endpoint ID

Device UUID

Hostname History

Hardware Fingerprint

Operating System Fingerprint

BIOS Information

Virtualization Status

Owner

Department

Environment

Nothing is guessed.

Everything is verified.

Chapter 2

Hardware Intelligence Engine

Responsible for

CPU

Memory

Storage

Motherboard

Firmware

TPM

Secure Boot

Virtualization

Battery (Laptop)

Peripheral Devices

Hardware changes become timeline events.

Chapter 3

Operating System Intelligence

Collects

Operating System Version

Kernel Version

Installed Updates

Security Patches

Update History

Build Number

Architecture

Support Status

End-of-Life Information

The engine tracks support lifecycles so administrators know when upgrades are required.

Chapter 4

Process Intelligence Engine

Rather than simply listing processes.

Every process becomes an object.

Example

```
Process
```

The platform understands relationships.

Chapter 5

Service Intelligence Engine

Collects

System Services

Startup Type

Privileges

Dependencies

Binary Location

Publisher

History

Unexpected Changes

Chapter 6

Startup Intelligence Engine

Observes legitimate persistence mechanisms provided by the operating system, such as:

- Startup folders.
- Scheduled tasks.
- Login items.
- Service autostart configuration.
- Registry-based autostart locations (Windows).
- Systemd units (Linux).
- LaunchAgents/LaunchDaemons (macOS).

The goal is to help users identify unexpected persistence **on systems they own or are authorized to manage**.

Chapter 7

File Intelligence Engine

Every important system file receives

Hash

Size

Signature Status

Version

Owner

Permissions

History

Trust

Baseline comparison highlights unexpected modifications.

Chapter 8

Software Intelligence Engine

Every installed application becomes an object.

Tracks

Version

Vendor

Installation Date

Update History

Dependencies

Digital Signature

Support Status

Known Security Advisories

License Information (optional)

Chapter 9

Configuration Intelligence Engine

Evaluates

Firewall Configuration

Disk Encryption

Secure Boot

Password Policies

Automatic Updates

Remote Access Settings

Audit Policies

System Logging

Configuration is compared with approved organizational baselines.

Chapter 10

Network Intelligence Engine

Observes

Network Interfaces

DNS Configuration

Gateway

VPN

Known Connections

Firewall State

Routing

Listening Services

The engine focuses on visibility and configuration assessment.

Chapter 11

Endpoint Trust Engine

Every endpoint receives independent trust dimensions.

Examples

Operating System Trust

Hardware Trust

Identity Trust

Application Trust

Configuration Trust

Network Trust

Overall Trust

Users can drill into every score to see the supporting evidence.

PS-ADR-0023

Endpoint as a Living Object

Status

Accepted

Decision

Endpoints shall be modeled as living objects with relationships, timelines, trust profiles, and evidence rather than static inventory records.

Rationale

This enables richer analysis, historical understanding, and better operational decisions.

NEW CORE ENGINE

Endpoint Baseline Engine

One of the most important engines.

Every endpoint establishes a verified baseline.

Includes

Approved Software

Approved Services

Approved Startup Items

Expected Configuration

Expected Hardware

Expected Policies

Future assessments compare against this baseline.

NEW CORE ENGINE

Endpoint Change Intelligence Engine

Rather than simply saying

> Something changed.

It answers

What changed?

When?

Who initiated it (where observable)?

Expected?

Approved?

Verified?

Impact?

Related Objects?

This greatly improves operational awareness.

NEW CORE ENGINE

Endpoint Health Engine

Health is measured across several dimensions.

Dimension	Example
OS Health	99%

NEW IDEA

Trusted Application Catalog

Organizations can maintain a signed catalog of approved software.

The platform compares observed software against the approved catalog and highlights:

- New applications.
- Retired applications.
- Unsupported versions.
- Missing signatures.
- Unexpected publishers.

This supports software governance without assuming that unknown software is automatically malicious.

NEW IDEA

Endpoint Story

Every endpoint has its own narrative.

Example

> "This laptop was enrolled on March 3. Full disk encryption was enabled on the same day. A new VPN client was installed in April. During June, two configuration changes reduced the Configuration Trust score. Both changes were verified and remediated within 24 hours."

Every sentence links back to evidence.

NEW IDEA

User-Friendly Endpoint Experience

One of your earliest requirements has always been:

> **"The platform must be easy enough for private users while remaining powerful for professionals."**

I think we should formalize that.

When a home user selects **"Check My Computer"**, they should **not** see 2,000 technical findings.

Instead, they should see something like:

Your Computer

Overall Health: Excellent (96%)

Protection: Active

Updates: Current

Recommendations:

- Enable firmware updates.
- Remove one application you no longer use.
- Review one application with extensive permissions.

Estimated time: **12 minutes**

"Show Technical Details" expands into the evidence, timelines, objects, and relationships for advanced users.

This is consistent with our mission-driven philosophy.

A critical design decision

Earlier in our discussions you asked whether Project Sentinel should be able to determine if a computer has been hacked or contains hidden applications.

I think we should phrase our design goal carefully:

Project Sentinel should help users detect signs of compromise, unexpected persistence mechanisms, unauthorized configuration changes, suspicious software, and deviations from trusted baselines on systems they own or are authorized to manage.

That is a realistic and technically achievable goal.

It avoids promising perfect detection—something no security product can honestly guarantee—while still aiming to provide deep visibility and actionable guidance.

Looking Ahead

Volume 27 will define the **Sentinel Mobile Intelligence Platform**, one of the most ambitious parts of the project because it must work across Android and iOS while respecting each operating system's security architecture.

Instead of trying to bypass platform protections, we will design it to:

- Use official platform APIs where available.
- Collect evidence with user authorization.
- Assess device health, application inventory, permissions, encryption, updates, and configuration.
- Detect deviations from trusted baselines.
- Integrate seamlessly into the same Object, Mission, Evidence, and Knowledge models as every other component.

That approach is technically feasible, maintainable, and suitable for individuals, enterprises, financial institutions, and government deployments. It also aligns with our guiding principle: **Project Sentinel earns trust by being transparent, evidence-driven, and engineered for the long term.**

=====[USER]=====

Can our system scan or check many computers/laptops at samme time if a company/organization want to check if the employee laptops/computers are safe and not has been compromissed and the same for mobil/ipad devices ? Good and continue

=====[ASSISTANT]=====

Absolutely.

In fact, **if Project Sentinel cannot do this efficiently, it will never be competitive in the enterprise market.**

What you are describing is one of the core reasons companies buy enterprise cybersecurity platforms:

> **"Show me the security posture of all my devices from one place."**

I think we should go **beyond** current products.

My vision

Project Sentinel should work equally well for:

- 1 private person
- A family with 12 devices
- A company with 500 laptops
- A bank with 50,000 endpoints
- Government agencies with hundreds of thousands of devices

The user experience should remain the same.

Only the scale changes.

PROJECT SENTINEL BLUEPRINT

Volume 27